

Lab 7

2DX3

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Claresta Cheung, Luke Xiao

400582489, 400559483

As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is our own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario.

1. When referring to interrupt triggering via GPIO, explain the difference between falling edge, rising edge, low level, and high level (Hint: see textbook section 9.5)

Falling edge is when the signal transitions from a high voltage (3.3V or 5V) to a low voltage (0V), triggering an interrupt.

Rising edge is when the signal transitions from a low voltage (0V) to a high voltage, triggering an interrupt.

Low level is when the voltage on the pin remains at 0V, causing the interrupt to trigger and continue to re-trigger.

High level is when the voltage on the pin remains high (3.3V or 5V), causing the interrupt to trigger and continue to re-trigger.

Falling edge and rising edge are edge/transition sensitive. Low level and high level are level sensitive.

2. What five conditions must be true for an interrupt to occur

1. The trigger flag bit is set (RIS)
2. The arm bit is set (IME)
3. The level of the edge-triggered interrupt must be less than BASEPRI
4. The edge-triggered interrupt must be enabled in the NVIC_EN0_R
5. Bit 0 of the special register PRIMASK is 0

3. How do you enable interrupts?

1. Arm the source of interrupt
2. Enable interrupt in the NVIC
3. Enable global interrupts
4. Set interrupt priority level
5. Write the interrupt service routine to handle the interrupt

4. What are the steps that occur when an interrupt is processed?

1. Registers (PSR, PC, LR, R12-R0) pushed on the software stack
2. Program Counter set to Vector Address
3. IPSR set to 18 (interrupt program status register)
 - a. Recall NVIC - see 18 refers to PortC
4. LR
 - a. LR's top 24 bits set to FFFFFFFF (indicating ISR)

- b. LR's bottom 8 bits set to (some #), indicating return method using the MSP

5. Use MSP as stack pointer